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Pharmacogenomic analysis summary

sample	NA12878
analysed	2026-10-14T18:53:41.752084Z
printed	2026-10-14
drugs checked	6

This is not a clinical report. It was produced by a final-year student project for research and education. It has not been clinically validated, and no qualified clinical expert has reviewed the explanatory text on this page.

Recommendation text on this page is quoted from published CPIC guidance via PharmCAT. Automated source checks are not human review: they cannot show that every statement here is supported, or that any of it is correct for the person this sample came from.

Do not change a medicine on the strength of this page. Take it to a qualified healthcare professional.

Results

Ordered by consequence, not alphabetically.

clopidogrel **Ineffective**

gene	CYP2C19
diplotype	*1/*2
phenotype	IM
evidence level	Unknown

clopidogrel: CYP2C19 *1/*2 (Intermediate Metabolizer) — Ineffective.

CPIC GUIDELINE — QUOTED EXACTLY

Avoid standard dose (75 mg) clopidogrel if possible. Use prasugrel or ticagrelor at standard dose if no contraindication.

CPIC strength of recommendation: Strong. Population: CVI ACS PCI. Implications: CYP2C19: Reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

— Annotation of CPIC Guideline for clopidogrel and CYP2C19 (via PharmCAT) [label rule: avoid_for_lack_of_efficacy]

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azathioprine **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

azathioprine: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

Required genomic evidence is incomplete for TPMT, NUDT15. Available partner findings are partial evidence only; no complete drug-level PGx recommendation is made.

Annotation of CPIC Guideline for azathioprine and NUDT15, TPMT

— PharmCAT (no matching CPIC entry)

codeine **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

codeine: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

CYP2D6 copy-number and structural variation is not called from the uploaded VCF in this application, so codeine guidance is not produced.

Annotation of CPIC Guideline for codeine and CYP2D6

— PharmCAT (no matching CPIC entry)

fluorouracil **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

fluorouracil: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

Required genomic evidence is incomplete for DPYD. Available partner findings are partial evidence only; no complete drug-level PGx recommendation is made.

Annotation of CPIC Guideline for fluorouracil and DPYD

— PharmCAT (no matching CPIC entry)

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simvastatin Unknown

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

simvastatin: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

Required genomic evidence is incomplete for SLCO1B1. Available partner findings are partial evidence only; no complete drug-level PGx recommendation is made.

Annotation of CPIC Guideline for simvastatin and SLCO1B1

— PharmCAT (no matching CPIC entry)

warfarin Unknown

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

warfarin: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

No validated automated warfarin-dose implementation is provided by this application; a gene call alone is not a complete dosing recommendation.

Annotation of CPIC Guideline for warfarin and CYP2C9, CYP4F2, VKORC1

— PharmCAT (no matching CPIC entry)

Unknown results (5)

An Unknown is a declined answer, not a missing one. Only the reasons about the file itself — missing positions or unusable genotypes — are something the sample provider can act on.

azathioprine Your file is missing required positions

Some of the positions needed to identify this gene carry no genotype in your file. That is not a small gap: a change whose key position is absent is invisible, so the genotype would read as normal and a reduced-function result could be reported as safe. The system declines rather than risk that.

What would fix it: Upload a VCF that reports ALL positions, including those matching the reference — a clinical pharmacogenomic panel, or data covering all your genes, produced with every position reported. A variants-only file cannot work here.

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codeine Not called from an uploaded VCF

This gene's result depends on structural and copy-number changes that need specialised analysis of the sequencing reads, which this application does not perform. No VCF uploaded here will resolve it — it needs a specialised test or caller. The system declines to guess rather than report a genotype it cannot support.

Nothing about the uploaded file would change this result.

fluorouracil Your file is missing required positions

Some of the positions needed to identify this gene carry no genotype in your file. That is not a small gap: a change whose key position is absent is invisible, so the genotype would read as normal and a reduced-function result could be reported as safe. The system declines rather than risk that.

What would fix it: Upload a VCF that reports ALL positions, including those matching the reference — a clinical pharmacogenomic panel, or data covering all your genes, produced with every position reported. A variants-only file cannot work here.

simvastatin Your file is missing required positions

Some of the positions needed to identify this gene carry no genotype in your file. That is not a small gap: a change whose key position is absent is invisible, so the genotype would read as normal and a reduced-function result could be reported as safe. The system declines rather than risk that.

What would fix it: Upload a VCF that reports ALL positions, including those matching the reference — a clinical pharmacogenomic panel, or data covering all your genes, produced with every position reported. A variants-only file cannot work here.

warfarin Result withheld

No validated automated warfarin-dose implementation is provided by this application; a gene call alone is not a complete dosing recommendation.

Nothing about the uploaded file would change this result.

Input coverage

How many of the positions each gene needs carried a usable genotype in the uploaded file. A genotype caller reads an absent position as matching the reference, so missing data can produce confident results rather than uncertain ones — which is why the count is printed whether or not it passed, and why a gene below its minimum is reported as Unknown.

gene	usable	required	minimum	sufficient
CYP2C19	35	35	100%	yes
CYP2C9	0	88	100%	NO
CYP2D6	0	157	100%	NO
DPYD	0	83	20%	NO
NUDT15	0	20	80%	NO
SLCO1B1	0	35	100%	NO
TPMT	0	45	80%	NO

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Not resolved from this file: CYP2C9, CYP2D6, DPYD, NUDT15, SLCO1B1, TPMT. Any drug keyed to these genes was reported as Unknown rather than assumed normal.

Where this guidance came from

PharmCAT	3.4.0
CPIC data version	2026-07-13-11-40
explanations generated	24 Jul 2026
source	CPIC guidelines, retrieved via PharmCAT

Guidance reflects what CPIC published when this data was captured. CPIC revises its guidelines; this build does not monitor for changes.